

Incoming SCADA Telemetry Stream

Inputs: Wind Speed v , Power P , Pitch β , Time t
Clean ($\tau = 0$) | Latency ($\tau \leq 20 \text{ m}$) | Drop ($\beta = \emptyset$)



Observability & Staleness Gate

Audits Pitch Register $\exists \beta$ and Latency τ

$\tau = 0, \exists \beta$

$\tau \leq 20 \text{ m}, \exists \beta$

$\beta = \emptyset$ or $\tau \geq 60 \text{ m}$

Tier 1: Physical

Continuous Quantile

- Nominal $\tau = 0$
- 6.8% violation
- Zero ML cost
- IEC 61400 rule

[Lowest Cost]

Tier 2: Recalibrate

State-Conditional

- $\tau \leq 20 \text{ min}$
- Pitch $\exists \beta$
- Absorbs 55% loss
- Empirical $q_\tau(v)$

[Recalibration]

Tier 3: STGQ-Mod

Residual Head

- $\beta = \emptyset$ or $\tau \geq 60 \text{ m}$
- 9.7% violation
- Saves 46k-68k
- Latent recovery

[Representation]